

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I N. Chawran Aryan(192365053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawran Aryan

Signature of the student:

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty
- ii. Explain how A* would handle the same situation, including how it balances cost and heuristic
- iii. Analyze the impact of heuristic accuracy on both algorithms
- iv. Discuss how dynamic changes (blocked paths) affect search performance
- v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

- i. Explain how Hill Climbing would work in this scenario and identify its limitations
- ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation
- iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations
- iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

- i. Explain why this problem requires an online search agent instead of offline search
- ii. Analyze the challenges of partial observability and dynamic environments

- iii. Describe how the rover builds and updates its internal model
- iv. Compare online search with uninformed and informed search strategies
- v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

- i. Clearly define variables, domains, and constraints with explanation
- ii. Explain how backtracking search works in this scenario
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency
- iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must compete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem
- ii. Analyze the computational challenges of large game trees
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning
- iv. Design an evaluation function and justify the factors considered
- v. Recommend a strategy for balancing optimality and real-time performance

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. AI-Powered Drone Delivery (Greedy Search & A*)

i. Greedy Best-First Search

Greedy Best-First Search selects the path with the **lowest heuristic value ($h(n)$)**, such as the shortest straight-line distance to the destination.

Strengths

- Fast route selection.
- Uses less memory.
- Suitable for emergency situations where quick decisions are required.

Weaknesses

- Ignores the actual travel cost.
- May choose blocked or risky paths.
- Does not guarantee the optimal route.

ii. A* Search

A* uses the evaluation function

$$[f(n) = g(n) + h(n)]$$

where:

- **$g(n)$** = actual cost from the start node.
- **$h(n)$** = estimated cost to the goal.

A* considers both the travelled cost and estimated remaining distance, allowing it to avoid costly or blocked routes while still finding an efficient path.

iii. Impact of Heuristic Accuracy

- **Accurate heuristic:** Faster search and optimal path.
- **Overestimated heuristic:** May produce non-optimal paths.
- **Poor heuristic:** Increases search time and explores more nodes.

iv. Effect of Dynamic Changes

When roads become blocked:

- Greedy Search may continue toward the blocked path and must restart.
- A* updates the path cost and replans efficiently using the new information.

v. Recommended Algorithm

A* is recommended because:

- Produces optimal routes.
- Considers both travel cost and heuristic.
- Handles changing environments better.
- Ensures safer and more reliable delivery.

2. Smart Traffic Signal Optimization (Hill Climbing & Genetic Algorithm)

Problem Formulation

Objective:

- Minimize waiting time.
- Minimize fuel consumption.
- Reduce traffic congestion.

Traditional search is inefficient because there are millions of possible signal timing combinations.

i. Hill Climbing

Hill Climbing starts with an initial signal timing and repeatedly changes it to improve traffic flow.

Advantages

- Simple.
- Fast.
- Low memory usage.

Limitations

- Can become trapped in local optimum.
- Cannot always find the global optimum.

ii. Local Maxima, Plateaus and Ridges

Local Maximum

- Algorithm finds a good traffic schedule but not the best one.

Plateau

- Several signal timings produce nearly identical performance.

Ridge

- Better solutions exist but require multiple simultaneous timing changes.

iii. Simulated Annealing / Genetic Algorithm

Simulated Annealing

- Occasionally accepts worse solutions.
- Escapes local maxima.

Genetic Algorithm

- Maintains multiple candidate solutions.
- Uses selection, crossover and mutation.
- Explores a larger search space.

iv. Recommended Method

Genetic Algorithm

Reason

- Highly scalable.
- Adapts to changing traffic.
- Produces near-optimal solutions for large cities.

3. Mars Rover (Online Search Agent)

i. Why Online Search?

The rover has no complete map of Mars.

It must:

- Explore.
- Sense.
- Learn.
- Decide simultaneously.

Therefore, an **Online Search Agent** is required.

ii. Challenges

- Partial observability.
- Unknown obstacles.
- Dynamic terrain.
- Limited battery.
- Communication delay

iii. Internal Model

The rover continuously updates:

- Visited locations.
- Hazards.
- Safe paths.
- Sample locations.
- Terrain information.

This updated map helps future decisions.

iv. Comparison

Online Search

Explores while moving

Suitable for unknown environments

Adaptive

Uninformed Search

Needs full map

Suitable for known environments

Not adaptive

Informed Search

Uses heuristic

Suitable for known environments

Partially adaptive

v. Recommended Approach

Online Search Agent

Reason:

- Learns continuously.
- Adapts to uncertainty.
- Makes safe real-time decisions.

4. University Exam Timetable (Constraint Satisfaction Problem)

Problem Formulation

Variables

- Exam subjects

Domain

- Available time slots
- Exam halls

Constraints

- No overlapping exams.
- Hall capacity must not exceed limit.
- Faculty availability.
- Subject ordering.
- Special accommodation.

i. Variables, Domains and Constraints

Variables:

- Each course exam.

Domains:

- Time slots and halls.

Constraints:

- One exam per hall at a time.
- Students cannot have two exams simultaneously.
- Faculty must be available.
- Hall capacity must be sufficient.

ii. Backtracking Search

Steps:

1. Assign a time slot.
2. Check constraints.
3. If violated, backtrack.
4. Try another assignment.
5. Continue until all exams are scheduled.

iii. Forward Checking

Forward checking:

- Removes invalid future assignments.
- Detects conflicts early.
- Reduces unnecessary backtracking.
- Improves efficiency.

iv. Best Strategy

Backtracking with Forward Checking and Constraint Propagation.

Reason:

- Faster.

- Scalable.
- Efficient for large universities.

5. Strategic Game AI (Minimax & Alpha-Beta Pruning)

i. Minimax Algorithm

Minimax assumes:

- AI tries to maximize its score.
- Opponent tries to minimize AI's score.

The algorithm explores the game tree and selects the move with the highest guaranteed score.

ii. Computational Challenges

- Huge game tree.
- High memory usage.
- Long computation time.
- Not suitable for real-time games without optimization.

iii. Alpha-Beta Pruning

Alpha-Beta Pruning eliminates branches that cannot influence the final decision.

Benefits

- Reduces node expansion.
- Faster search.
- Same optimal result as Minimax.
- Enables deeper search within time limits.

iv. Evaluation Function

The evaluation function considers:

- Player health.
- Available resources.
- Number of units.
- Territory control.
- Distance to enemy.
- Attack and defense strength.

A higher evaluation score indicates a stronger game position.

v. Recommended Strategy

Minimax with Alpha-Beta Pruning

Justification

- Produces optimal decisions.
- Reduces computation time.
- Suitable for real-time games.
- Efficiently searches deeper game trees.